

IObjectType SCRIPT INTERFACE

The `IObjectType` script interface represents an object type in the project, e.g. a `Sprite` object type. An object type can have multiple instances created, which are represented by the `IWorldInstance` script interface if it appears in a layout, otherwise the `IInstance` interface.

`IObjectType` derives from `IObjectClass` which is a base class shared with families (`IFamily`).

Getting an IObjectType

References to the project's object types are typically accessed through the `IRuntime` interface `objects` property. For example `runtime.objects.Sprite` would refer to the `IObjectType` interface for the *Sprite* object type, assuming one was added to the project.

⚠ Try not to confuse object classes with object instances. A common mistake is to try to use something like `runtime.objects.Sprite.x` to get the X co-ordinate of a `Sprite` instance. However `runtime.objects.Sprite` is an `IObjectType`, which does not have a position. First add another call to get an instance before trying to read instance properties, for example `runtime.objects.Sprite.getFirstInstance().x`.

Examples

The following examples demonstrate using some features of `IObjectType`:

- **Instance destroy event** demonstrates the use of the `"instancedestroy"` event

- [Iterating instances](#) demonstrates the use of the `instances()` iterator to modify all instances

Object type APIs

`setInstanceClass(Class)`

Set a custom class to be used to represent instances of this object type. The class must derive from the default class. This can only be called in `runOnStartup`, before any instances have been created. For more information see the guide on [subclassing instances](#).

`createInstance(layerNameOrIndex, x, y, createHierarchy, template)`

Create a new instance of the object type at a position. The layer to create on is specified either by a case-insensitive string of the layer name or its zero-based index. The position is given in layout co-ordinates. If `createHierarchy` is true, all children of the created instance in the scene-graph hierarchy will also be created automatically with their connections in place. If `template` is a valid template name then the new instance will be based on the template rather than an arbitrary instance. The return value is the script interface for the created instance.

⚠ See *Setting up a hierarchy* in the [Layout View manual entry](#) for more information about hierarchies.

⚠ See the [Templates manual entry](#) for more information on what templates are and how to start using them.

`getAllFamilies()`

Return an array of [IFamily](#) listing all the families this object type belongs to.

`*families()`

Iterates over all the families this object type belongs to (as `IFamily`).

`isInFamily(family)`

Returns a boolean indicating if this object type belongs to the specified `IFamily`.

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This manual entry was last updated on 21 May, 2025 at 13:00